DigiPick of the month

Booting Up The Net!

This month's winner is

Subhash Bhojwani
e-mail: mirashub@gmail.com

He wins

Visual Basic Database Programming

by Roger Jennings
Published by
WILEY-INDIA

WIN!
Send in your entry and you could win an exciting gift just by sharing an amusing picture with a tech angle to it. The picture should have been shot by you, and should not have been published anywhere earlier. E-mail your picture with the subject **'DigiPick'** and your postal address **on or before the 15th of this month** to **digipick@thinkdigit.com.** One prize-winning picture will be published each month.

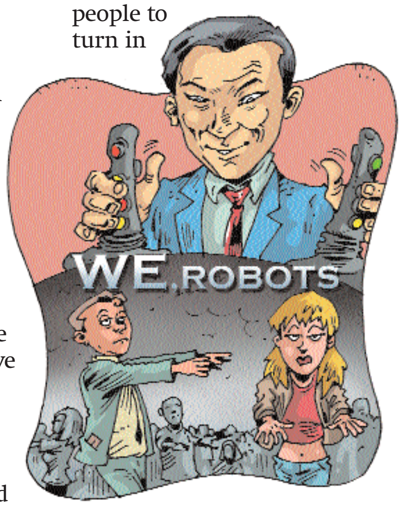
HUMAN ROBOTS

The Jedi Knights Of Japan

Yes, perhaps that's what we should call Japanese company Nippon Telegraph and Telephone (better known as NTT), because they've come up with a way to make you do things you don't really want to! Though not as easily as the Jedi do in the *Star Wars* saga (with a wave of the hand), NTT has made a headset that can force a wearer to move against his or her will! The commands are passed via a pair of remote-controlled joysticks.

The device uses "galvanic vestibular stimulation" (GVS); it basically controls the wearer's balance via their ears. The headset, which looks like a pair of large headphones, fit over the subject's ears, and balance is controlled using electrical impulses that are passed to the nerves.

Apart from forcing people to turn in



a specified direction when walking, the device also simulates the feeling of turning, even if the subject is motionless—sort of like virtual reality. Thus, the device can even trick the wearer into feeling motion that isn't really happening.

Enough about the device though: what we're more concerned about is why someone would want to create such a technology!

Though the researchers claim they've developed the device to "enhance" a gamer's experience, we're still at a loss for why a gamer would need his or her mind controlled. Sure you could simulate more surreal motion, but are we really ready to allow games to control any part of our minds (or even ears)?

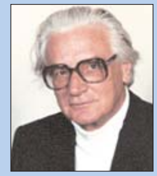
Thankfully, the device is still not something you can just go out and buy! Because it works by directly applying currents to nerves in a subject's head, it's dangerous, and if wrongly used, could cause burning of the skin or nerve damage—not something a lot of people are willing to risk just to have a more realistic gaming experience.

The technology could advance, going beyond having to attach a headset—GVS opens up a whole range of possibilities. It's the beginnings of mind control. One shudders to think about the possibilities were GVS technology to become mainstream: imagine police using it to quickly disperse crowds, or teens being forced to return directly home from school. Is this how we want to end up? Controlled by the people who can afford the technology? Let's hope there are enough upholders of human rights and ethics to cry foul when attempts to commercialise such technology finally crop up. Some technologies are better left undiscovered!

People Who Changed Computing

The Original Computor

It's strange that we've heard so little about Konrad Zuse, a German engineer, who probably invented the computer. "Probably," because there are several definitions of "computer," and one of his machines—the Z3—has a strong claim to being the first computer.



Konrad Zuse

Formally, the Z3 was the first (functional) stored-program-controlled computer. The Z3 was preceded by the Z2 and the Z1, which was a binary electrically-driven mechanical calculator. All three were destroyed in WWII.

Zuse also founded, in 1946, the first-ever computer startup company, the Zuse-Ingenieurbüro Hopferau. More firsts: he founded another company, Zuse KG, in 1949, which created the Z4 in 1950. At the time, it was the only working computer in Europe excluding Britain, and the first computer in the world to have been sold! One wonders how much more Zuse might have achieved had he been born in a country such as Britain or the US during the War, instead of in Germany. Or if there had been no War at all.

Zuse suggested, as a scientific philosophy, that the universe is running on a grid of computers. After he retired, he concentrated on painting, a long-time hobby. He died in 1995.

In the words of his son Prof Horst Zuse, "Many reference works state that the first large-scale automatic digital computer was the Harvard Mark 1... However, the Z3 pre-dated the Harvard Mark I. Today, Konrad Zuse is acclaimed... as being the most admired and respected computer pioneer." Visit www.epemag.com/zuse for a biography of Zuse by his son.